

Task 1: Install Apache and Verify Installation

Install Apache (httpd)

- `sudo dnf install -y httpd`

Enable and Start the Apache Service

- `sudo systemctl enable --now httpd`

Check the service status:

- `sudo systemctl status httpd`

Allow HTTP Through the Firewall

- `sudo firewall-cmd --permanent --add-service=http`
- `sudo firewall-cmd --reload`

Test Apache

- `http://<Server-IP>`

Task 2: Change the Default Homepage

Open the default webpage:

- `sudo nano /var/www/html/index.html`

Replace its contents with:

```
<!DOCTYPE html>
<html>
<head>
  <title>My Apache Server</title>
</head>
<body>
  <h1>Welcome to My Apache Web Server</h1>

  <h2>Name: Nishit Panchal</h2>

  <h2>Favorite IPL Team: Gujarat Titans</h2>
</body>
</html>
```

Restart Apache:

- `sudo systemctl restart httpd`

Open:

- `http://localhost`

Task 3: Create a Virtual Host

Create the website directory:

- `sudo mkdir -p /var/www/myzomatoclone`

Create the homepage:

- `sudo nano /var/www/myzomatoclone/index.html`

Set permissions:

- `sudo chown -R apache:apache /var/www/myzomatoclone`
- `sudo chmod -R 755 /var/www/myzomatoclone`

Create the Virtual Host configuration:

- `sudo nano /etc/httpd/conf.d/myzomatoclone.conf`

Add:

```
<VirtualHost *:80>
  ServerName myzomatoclone.local
  DocumentRoot /var/www/myzomatoclone
  <Directory /var/www/myzomatoclone>
    AllowOverride All
    Require all granted
  </Directory>
  ErrorLog logs/myzomatoclone_error.log
  CustomLog logs/myzomatoclone_access.log combined
</VirtualHost>
```

Test the configuration:

- `sudo apachectl configtest`

Restart Apache:

- `sudo systemctl restart httpd`

Update the hosts file:

- `sudo nano /etc/hosts`

Add:

- `127.0.0.1 myzomatoclone.local`

Test in the browser:

- `http://myzomatoclone.local`

Task 4: Enable URL Rewriting

Verify mod_rewrite is loaded

- `httpd -M | grep rewrite`

Create a .htaccess file:

- `sudo nano /var/www/myzomatoclone/.htaccess`

Add:

`RewriteEngine On`

`RewriteRule ^foodie$ /index.html [L]`

Allow .htaccess by editing the virtual host:

- `sudo nano /etc/httpd/conf.d/myzomatoclone.conf`

Ensure the directory block contains:

```
<Directory /var/www/myzomatoclone>  
    AllowOverride All  
    Require all granted  
</Directory>
```

Restart Apache:

- `sudo systemctl restart httpd`

Test in the browser:

- `http://myzomatoclone.local/foodie`